

Perspectives of Assistive Technology Industry Professionals on AT for Children

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ABSTRACT

Assistive technology (AT) can help children with disabilities improve their learning outcomes, social skills, and autonomy, but past work has shown that AT is often abandoned or underutilized in classrooms and at home. However, little research focuses on the perspectives of the AT industry professionals who are supporting children, even though they often provide direct and indirect support to their customers. As such, we conducted semi-structured interviews with 21 AT industry professionals to investigate how AT companies are involved in the successful integration of AT into a child's life. We uncovered three strong needs: (1) improving communication channels between AT industry and support networks, (2) demonstrating the efficacy of AT and (3) exploring long term support of AT.

CCS CONCEPTS

• Human-centered computing Accessibility Empirical studies in accessibility;

KEYWORDS

accessibility, assistive technology, children with disabilities

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1 INTRODUCTION

Assistive technology (AT) can be essential to help children with disabilities access educational materials, communicate with others, be independent, and use technology [8, 15]. In the United States, when children receive AT as part of their Individualized Education Plan (IEP), they receive AT services “that directly assist a child

with a disability in the selection, acquisition, or use of an assistive technology” [40]. However, prior work has shown that children and their *supporters* - families, teachers, and specialists - face many challenges with these services, and AT is often abandoned or underutilized in classrooms and at home [4, 30, 33]. Although researchers have investigated the perspectives of the supporters [2, 9, 14, 41], there is less work on the role of AT companies, despite their critical involvement in these services.

AT companies play a part in children's AT services by marketing their products and services, helping people select AT, and providing initial training. They also have a business incentive to see the success of their products. Thus, *AT industry professionals*, individuals who work at AT companies and directly or indirectly work with children and their supporters, may play a critical role in children's success with AT. HCI researchers have asserted that involving all players relevant in AT is critical to seeing the impact of our research [7, 13, 23].

To fill the gap in the literature we explored the role that AT companies play in helping children and their supporters in helping children successfully use AT. We asked the following research questions: (1) What are the AT industry professionals' roles in the AT adoption process for children with disabilities? (2) What are the factors that impact the successful integration of AT into a child's life from the perspectives of AT industry professionals?

To answer these questions, we conducted semi-structured interviews with 21 AT industry professionals. These included both company representatives for companies that create or sell AT or who work with specialists to help select AT at an AT conference in the United States. We found that AT industry professionals' involvement throughout the AT adoption process depends on the type of AT, the knowledge and funding of a child and their supporters, and the way that AT is selected. Companies were directly involved with helping children and their support network select and initially learn to use AT, but most participants felt that children's long-term success with AT was largely out of their control.

Our contributions include three design opportunities to bridge the gap between industry and research: improving communication channels, demonstrating the efficacy of AT, and investigating long term support of AT.

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2 RELATED WORK

In the sections below, we present related work on AT investigating the perspectives of children, AT companies, and supporters (specialists and teachers). Our research builds on this prior work by including perspectives of AT industry professionals, which we found to be largely missing from the literature.

2.1 Assistive Technology and Children

Many researchers have worked to design and study new AT for children with disabilities. Specifically, they have investigated how AT can help children with disabilities improve learning outcomes [1, 10], communication abilities [5, 33, 35], executive functioning skills [43, 47], social skills [27], and daily living routines [3, 16]. Studies have also focused on the experiences and perceptions of children with disabilities, finding that children's perceptions of AT vary based on the nature of their environments and desired level of independence [18, 31], but children often face issues with ease of use of the technology and social implications [17].

Despite the positive findings cited above, researchers have found that AT often has high abandonment rates and issues along the integration process often arise [9, 29]. For example, abandonment rate for augmentative and alternative communication (AAC) devices range from 30 to 60%, due to high costs of the devices, complex programming, limited time for individuals to get familiar with the device before purchase, and slow communication rate [20, 38, 42]. More research is needed to understand the complete AT adoption process.

2.2 Perspectives of AT Industry Professionals

Although AT industry professionals have a vested interest in the successful implementation of AT in a child's life, we found little research examining their perspectives. Some of the prior research has focused on the relationships between AT companies and the users and specialists. Interviews by Danemayer et al. found that there was a disconnect between AT industry professionals and the users and that finding information about AT often required being a part of the AT industry [11]. However, there are legal and ethical considerations in the support and information that assistive technology companies can and cannot provide specialists in assisting with the selection and acquisition process [32]. Our research builds on this work, looking specifically at how AT industry professionals play a role in the different phases of the AT adoption process and the challenges from their perspective.

2.3 Perspectives of Teachers and Specialists

As there was little literature on the AT industry professional roles, we also looked at other roles that support children's use of AT, including the general and special education teachers and specialists (e.g. Teachers of the Visually Impaired, Occupational Therapists, etc.). Teachers can play an important role in the AT adoption process as many students use AT in the classroom, but face many challenges [15, 19, 25, 26, 40]. Specialists make the decision about which of AT to select [40], help students become familiar with the AT [22], train teachers and other school staff on how to integrate the AT into a student's daily school activities, and help families learn to use AT [30, 37].

Both types of supporters faced challenges in supporting AT use. Teachers are not always provided with appropriate AT to use with their students [46] and have difficulty acquiring and using them [44]. In addition to a lack of knowledge of AT, researchers have found that teachers face technical issues, have a lack of confidence, and develop negative attitudes [34]. Additionally, teachers are not always provided with training on the basics of AT or ongoing support on how to use AT [14, 24]. Compared with families and teachers, specialists have the most training in AT, but may spend the least amount of time with children. Specialists may not have enough knowledge to conduct appropriate evaluations and families may not be able to afford enough help from specialists [9, 12, 14, 28, 37, 41]. Studies focusing on the perspectives of SLPs and TVIs found that a lack of training and ongoing support around the AT [6, 39, 45] and problems around device mismatch [21] contribute to device abandonment.

In summary, although they may have more training in AT through professional development than families, teachers and specialists face challenges supporting students with AT due to lack of knowledge, training and support on the AT devices. Teachers and specialists may interact with AT industry professionals for professional training and at conferences; however, these interactions have been underexplored in the literature. Our work examines these interactions from an AT industry professionals' perspective.

3 METHODS

To better understand how AT companies are involved in the AT adoption process, we conducted semi-structured interviews with 21 representatives from AT companies. We then transcribed and analyzed the interviews to identify relevant themes for each phase involved in implementation of AT. All study procedures had institutional IRB approval.

3.1 Participants

Participants were selected from the company representatives who were present at a large US Assistive Technology conference in 2023. Over 70 organizations that support people with disabilities had exhibitor booths at the conference. We selected companies that provide AT for children with disabilities aged 3-12, and we selected participants from each company that could speak about their interaction with children.

Twenty-one individuals from seventeen unique companies participated in the interviews (Table 1). Interview participants both directly or indirectly supported children who use AT. For example, indirect supporters include company CEOs, Director of Strategic Partnerships, and Marketing Managers, who manage company operations that support children. Direct supporters include consultants that actively work with children to help evaluate whether the AT that the company provides is suitable for a child or help train individuals to use AT.

3.2 Procedure

All interviews were conducted during the conference with conference exhibitors. During the conference, a researcher approached the exhibit booth from one of the selected companies and asked

Table 1: Participant Summary. Company provides an overview of company they work for.

ID	Company	ID	Company
P1	Creates and sells AAC & alternative input software/hardware	P11	Creates and sells cameras to aid reading
P2	Training on & reselling AT devices	P12	Eye-tracking software and hardware
P3	Creates and sells AAC devices	P13	Sells text-to-speech pens
P4	Creates and sells AAC devices	P14	Creates hardware for vision
P5A	Sells AAC Devices, perform AT evaluations	P15	Creates and sells AAC & other alternative input software and hardware
P5B	Sells AAC Devices, perform AT evaluations	P16	Sells AT for vision
P6	Sells AAC Devices	P17	Reading comprehension software
P7	Creates hardware for vision	P18	Reading comprehension software
P8	Creates AAC accessories	P19	Digital curriculum
P9	Supports DIY AT	P20	AAC Software
P10	Accessories for iPad/AAC for motor/vision		

the representatives about their roles at the company. If a representative met the inclusion criteria, then the researchers asked the representative if they were interested in participating in the interview and briefly described the study. Interviews ranged in length from 15 to 25 minutes and were audio-recorded. Participants were compensated with a \$10 Amazon gift card for their time.

Interviews followed a semi-structured format and were organized along the following topics: 1) Background about the individual and company (e.g. What is your background with AT? How do your company support children with disabilities), 2) Understanding the support network (e.g. How does your company interact with parents, teachers, and therapists that work directly with children with physical disabilities?) and 3) Understanding each phase of the AT implementation process: selecting appropriate AT, acquiring chosen AT, and learning how to use AT (e.g. What is your involvement in the phase? What challenges do you or other individuals face? What practices have been most effective?).

3.3 Analysis

The audio-recorded interviews were transcribed to text files using professional software. We used open descriptive codes [36] to analyze the interviews and then draw insights. Three researchers brainstormed and developed an initial set of codes for the interviews. After independently coding several interviews, the researchers met to discuss any changes to codes and revised the codes together. Finally, all interviews were double coded by two researchers. When the researchers disagreed on a code, they discussed the differences until reaching agreement.

Codes were organized across several categories. We coded what part of the process of acquiring and learning they were discussing (e.g. selection, learn, etc.). We also coded various themes that arose around the barriers that were found (e.g. stigma, policy, etc.) and themes that encouraged successful use (independence, motivation, etc.).

4 FINDINGS

4.1 Marketing

Many participants noted that most of the time, families were unaware that AT exists and saw a need for AT companies to do more

marketing to get the word out about their products. P6 shared their opinion: “Not a lot of people know about it. I have my son who is autistic actually lives with a girl who is nonverbal, and they’ve never heard of AAC”, and P12: “This technology is so rare and niche that you don’t know it exists until you use it.” Children and their supporters may be informally introduced to AT through AT companies that hold workshops at their school (P9) or companies’ marketing efforts. Parents may also learn about AT anecdotally from other parents (P11) or as specialists may recommend AT use.

Participants discussed how marketing AT can differ from other technology. Since AT is usually purchased by schools and insurance, most marketing efforts were directed towards schools and professionals. P19 explained their company’s approach: “well, so children don’t buy directly, we come to conferences like this to present to teachers. And we also reach out to school districts around the United States...” Participants felt that the high price and small market of AT made marketing to schools challenging.

In addition to marketing directly to schools, participants said they targeted specialists, such as SLPs and OTs. P11 said “events like the [conference] are important for us because we have a lot of occupational therapists coming over” who use AT with children. P1 added that it was important for specialists to be aware of AT options and understand “what features and accessories are out there” to help with the selection process. P13 added that their company’s marketing budget was very small.

Three participants said their companies tried to follow a universal design approach, so they could sell their AT to a larger audience. P17 explained their reasoning: “I would say almost 50% of our customers look at this universally...it makes sense economically, from a pricing perspective” Although participants felt that marketing was important, many said that word of mouth was their primary acquisition method. P11 felt that if a child has success with AT, then “it just spreads around; as cliché as it sounds, [word of mouth] is really the best match.”

4.2 Selection and Trialing Process.

We found AT Industry professionals were very involved in helping children select AT. For AAC devices, participants discussed the

importance of feature matching between the AT and the child. Participants felt that making a sale was not their only goal, but also to make sure that customers were getting the best AT based on a child's needs. For example, P6 said "if our device doesn't work best... we won't recommend it." This sentiment was echoed by P14: "When somebody comes up to me . . . and says, 'this is wonderful, I really want one of these,' I'll say 'why'... With assistive technology, it needs to serve a purpose."

Participants analyzed a variety of data about a child and their environment to identify the best AT. P1 explained their process: "it's kind of a process of elimination. You're getting all this information about their physical abilities, their cognitive abilities, their linguistic abilities, and then trying to narrow down the options based on what features [the AT] has." P15 added "it's about how they learn how they think, how they handle the device when they get frustrated." Others also tried to understand the environment in which children will be using their AT including what tasks the child will be doing in the classroom (P2) and at home (P12). Participants felt the selection process was highly individualized. P15 described their perspective: "It really is an art, as well as a science. Every individual is different. . . . I wish there was a spreadsheet where we can type in a few things, and we can get the ideal system for each individual. But it's much more individualized than that."

As part of this complicated selection process, it was not unusual for multiple AT to be considered as the best match. Participants said that children received trials from a number of sources including directly from companies, through a tech lending library, or from borrowing from other school districts, and this trialing was common for selecting AAC devices, and even required by some insurance (P6). P20 explained that their goal was to identify several potential AAC devices for a child and then "see it in practice," during a trial period, before making a final decision. During the trial, participants worked directly with children and their supporters to evaluate whether the AT was an appropriate fit.

4.3 Funding Process Support.

AT is typically funded through insurance or school budgets. We found that AT companies were most involved in the funding process when AT could be covered by insurance, such as AAC devices. In order for an AAC to be covered through insurance, extensive requirements had to be met, which included a prescription from a physician and a formal evaluation process with an SLP. P5A indicated that it was a complicated process that can take 6 to 9 months. P4's company provided many resources to support this process such as an online course explaining the funding process and template documents that are used for evaluations. Unfortunately, even with the support that companies provided for the insurance process, there were still cases where insurance did not agree to fund the devices. P1 explained "So a lot of times insurance will come back to us. And they will say that it's not an evidence-based approach to that identified need."

4.4 Teaching AT

We found that industry professional's involvement in teaching AT was impacted by (1) the complexity of their AT, (2) children and supporters' prior knowledge with AT, and (3) company policies. P12

expressed that AT can be complex as it doesn't have any built-in tutorials. According to P4, "training and set-up usually takes about four hours, but it can take a lot less time if the SLP already knows what they are doing." However, there were some cases where the AT may be simple enough to learn with very minimal help (P10, P14, P17).

Many of the companies were involved with getting started with the newly acquired AT. Training usually involved setting up the AT and going over the basics of how to use it. Participants preferred to train everyone at the same time. P5 suggested that the best way to do training "is to go into the classroom, so that we get the teacher and the paraprofessional, and we invite the parents to come in which sometimes has to be done through a zoom type session." As families were often intimidated by AT, participants tried to help families "feel excited about what this technology can do and what doors they can open" (P3). P18 also explained their company's training strategy for engaging everyone at a school: "I feel like our best approach is when we train teachers and students simultaneously. When we train the teachers first, they do not feel comfortable when kids ask questions. But if you train them together, it really sets up their role better."

Depending on how AT was acquired, children received varying support from AT companies. P11 shared that if an AAC device was purchased directly from their company, they would provide complementary training, and "we don't leave until we know that person feels comfortable knowing the basics." However, if AT was acquired through external sources, such as receiving a second-hand device, then getting one-on-one support would be through a school specialist or an externally paid individual (P14, P10). After the initial training, companies offered varying levels of continued support. P1 said she would develop personal connections with families so they can "call me and ask me a little question." P16 said "we try to offer really good customer service as we find that it is critical for success." P20 explained how his company provided families with support after sessions covered by insurance run out: "there's a formal set of times that we contact a person to make sure that they're getting at least the most basic support possible."

Companies also offered indirect support through video training, digital or print manuals, webinars, and other resources. Participants explained that people have different learning preferences and availability, which they tried to accommodate (P10, P12). P20 described the educational methods that their company offered, beyond one-on-one training: "We have a host of resources that are posted on a website. And we rely on events like [AT conferences], or we have webinars. . . . We have beginner, intermediate, or expert courses that can be taken." P5 explained that every AAC company offers their own set of video tutorials that cover a range of content from the selection to long term use of AT, but several companies are well known for having great videos that can apply to AACs purchased from other companies. Participants also felt that short training videos could be more effective than long live or recorded sessions. P18 said their company made short two-minute training videos so "teachers could watch them during the first five minutes of their day".

4.5 Data Collection

Although we did not explicitly ask participants about collecting data around the AT use, we found that participants discussed the need to set and track goals relevant to children's progress with learning and using AT. Sometimes, AT companies were directly involved in helping supporters set goals. P19 suggested that the simplest way to track progress was with the AT itself: children use the AT and "we feed data back to the teacher about how they're doing."

Some companies conducted research studies to evaluate the impact of their AT. P17 said his company analyzed usage data for their "data-driven tools", and P19 said his company had done user interviews. P18 explained that her company had done several "impact studies" to show the long-term impact of their AT on students' reading abilities: "We have a data driven tool that we can actually measure the success with." In contrast, several participants noted that their companies had no direct involvement in tracking and data collection, which created a disconnect for the company in understanding the children's success in using the AT. For example, P5 said that the responsibility for tracking children's success with AT was "out of our hands; now it's into the hands of caregivers who work directly with the child" (P5). P11 explained that they only learned if AAC is working for a child through "word of mouth." P4 added that "it's too hard to know who is using [our AT] and for how long." P7 shared that "I think we need to start collecting data so that we can make educated decisions, both from a software development perspective, but also to benefit the end user."

5 DISCUSSION

Research has often focus on examining the usability of a single piece of AT with a small group of users, but our work has highlighted the importance of researching the entire AT adoption process and all players involved, including industry professionals. Below we discuss three areas where collaboration between researchers and industry could help.

5.1 Improving Communication Channels Between AT Industry and Support Networks

As supported by prior work and our findings, a critical challenge throughout the entire AT adoption process is the lack of knowledge about AT, affecting supporters' abilities to select appropriate AT and help their children use it [4, 15, 39, 43]. However, our participants added that lack of knowledge also makes it difficult for (1) children to be appropriately evaluated as children's supporters may not be aware that AT can help their child; (2) people to understand the importance of AT and not have a negative stigma; and (3) comply with institutional policies for acquiring AT that require proof of effectiveness. Through their company roles and past experiences working directly with children, AT industry professionals gain a great amount of knowledge about AT. We suggest that researchers explore how AT companies can share more of their knowledge with children and their supporters during all phases of the AT adoption process. Ideally, AT companies would provide more direct support, but we recognize that limited time and money makes this difficult. Given recent advancements in artificial intelligence (AI),

we suggest that researchers explore how AI can help with AAC feature matching, personalized training plans, and technical issues.

5.2 Demonstrating the Efficacy of AT

In our interviews, we uncovered a critical need for demonstrating the efficacy of AT, aligning with previous work by Danemayer et al. [11]. Sometimes families would select an AT only to have insurance companies reject their selection based on the unproven efficacy of the tool. For expensive AT, families might have to choose an alternative that might be less effective for their child. Additionally, there is a need to collect more data about children and how they use the AT to understand which configurations and which AT might work best for each child. This data could be used to support both selection (through better feature matching) and acquisition (through fewer roadblocks from insurance companies). Thus, we suggest that researchers help companies collect and analyze data to demonstrate the efficacy of existing AT.

5.3 Exploring Long Term Support of AT

Although some companies tracked the long-term use of their AT and found it beneficial, many participants noted that this was something their company did not currently focus on. Instead, these AT companies were mainly focused on helping families and school staff with the logistics of purchasing and setting up AT, and had collected little data about how children used the devices in the long-term. Therefore, in addition to helping demonstrate the effectiveness of AT, we suggest that researchers explore the factors that lead to long term success with AT, which could result in less abandonment of devices.

5.4 Limitations

Our participants were recruited at an AT conference in the United States, which impacted our results in a number of ways. Although we tried to interview participants who worked with a wide range of AT (Table 1), we found that there was a strong focus in our discussions on AAC and speech-based disabilities. In addition, many of the policies and laws discussed by the participants are specific to the US. Finally, the conference was primarily a networking and marketing opportunity, which may have affected their responses. Future work should explore more in-depth perspectives of industry professionals.

6 CONCLUSION

Our interview study addressed an important gap in the literature about the AT industry and the challenges and opportunities of successful AT integration for children with disabilities. We investigated the perspectives of AT industry professionals, who play a critical role in the AT adoption process and whose perspectives have rarely been explored. We discussed three ways researchers can help children by bridging the gap between industry and research: (1) improving communication channels, (2) demonstrating the efficacy of AT, and (3) investigating long term support of AT.

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